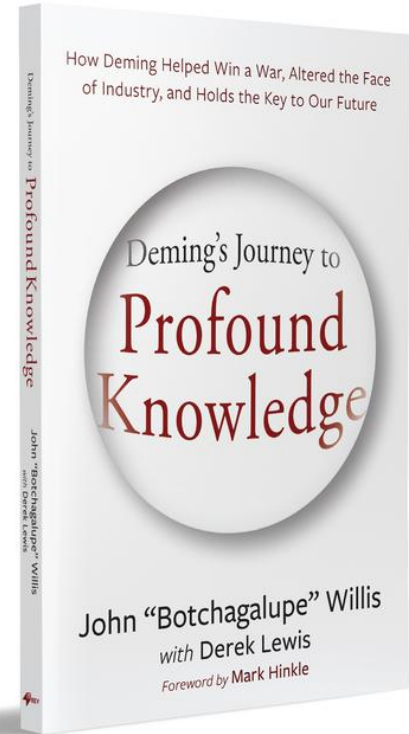


Beyond Checkboxes: Embedding Resilience

John Willis
@botchagalupe



Agenda

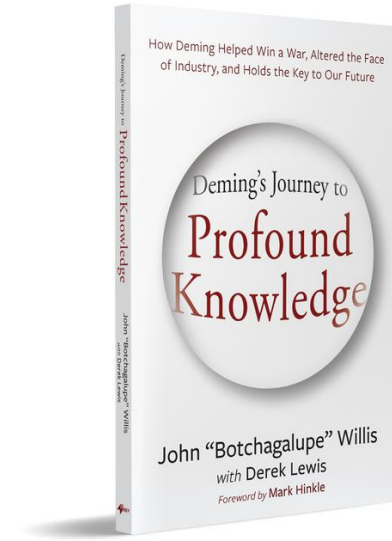
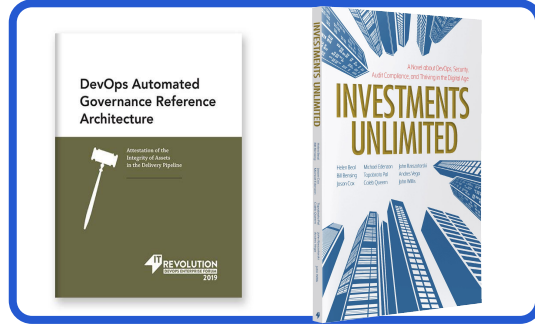
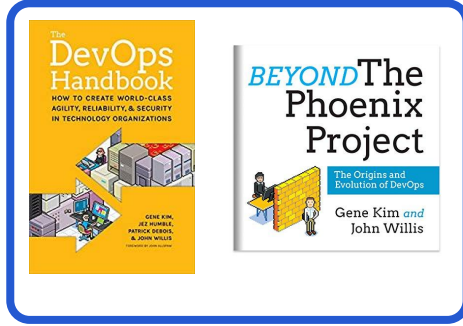
- Introduction
- Think Differently
- Who is Deming?
- Why Deming Matters?
- Why Now?



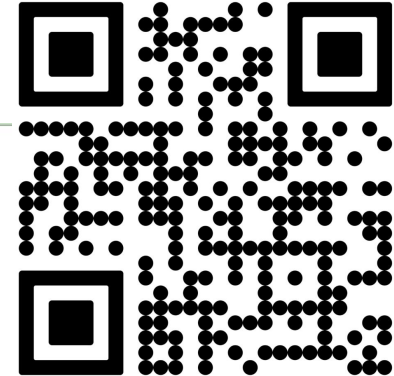
Introduction



@botchagalupe



John Willis  @botchagalupe

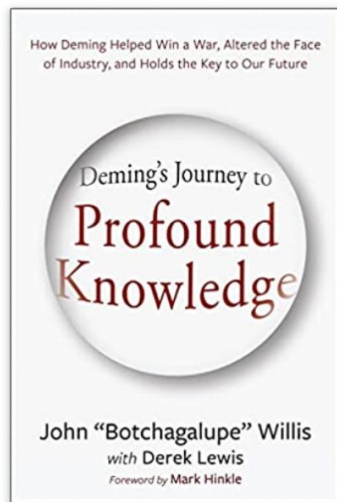




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Deming's Journey to Profound Knowledge: How Deming Helped Win a War, Altered the Face of Industry, and Holds the Key to Our Future Paperback – January 1, 2024



by [John Willis](#) (Author), [Derek Lewis](#)

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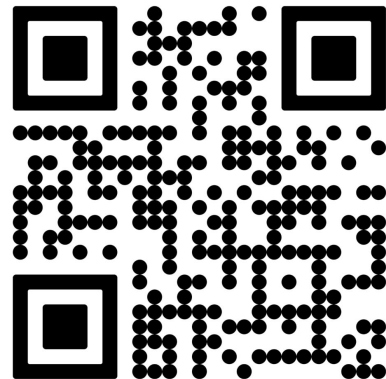
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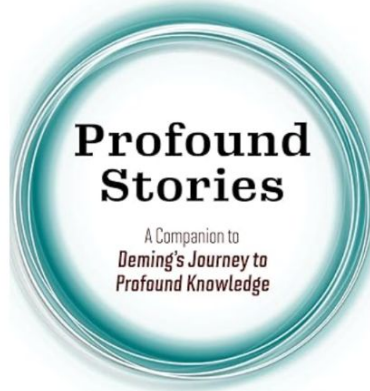
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"Read [Deming's Journey to Profound Knowledge] and you will understand like never before the wisdom and sound advice of Dr. Deming."
—Jeffrey K. Liker, author of *The Toyota Way*



JOHN WILLIS and DEREK LEWIS

Roll over image to zoom in

Profound Stories: A Companion to Deming's Journey to Profound Knowledge Kindle Edition

by [John Willis](#) (Author), [Derek Lewis](#) (Author) | Format: Kindle Edition

[See all formats and editions](#)

In this captivating companion to Deming's Journey to Profound Knowledge, authors John Willis and Derek Lewis share the untold stories and fascinating details that didn't make it into the original book. Profound Stories takes readers on a deeper dive into the life and times of W. Edwards Deming, offering rare insights and anecdotes that further illuminate the legendary figure's journey to developing his influential System of Profound Knowledge.

From Deming's humble origins to his wartime efforts and his pivotal role in Japan's post-war economic miracle, Willis and Lewis leave no stone unturned. Readers will discover the intriguing history behind key concepts like the PDSA Cycle and the Red Bead Game, as well as Deming's connections to other notable figures like Claude Shannon and Vannevar Bush.

Profound Stories and Deming's Journey to Profound Knowledge explores the far-



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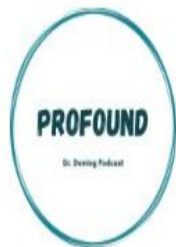
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NEWSLETTER

Profound

Profound is a newsletter applying W. Edwards Deming's ideas in the digital transformation era.



By John Willis

As an accomplished author and innovative entrepre...

Published weekly

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NEWSLETTER

Attention Is All You Need

A Botchagalupe spin on all thing Generative AI.



By John Willis

As an accomplished author and innovative entrepre...

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778 subscribers

Think Differently

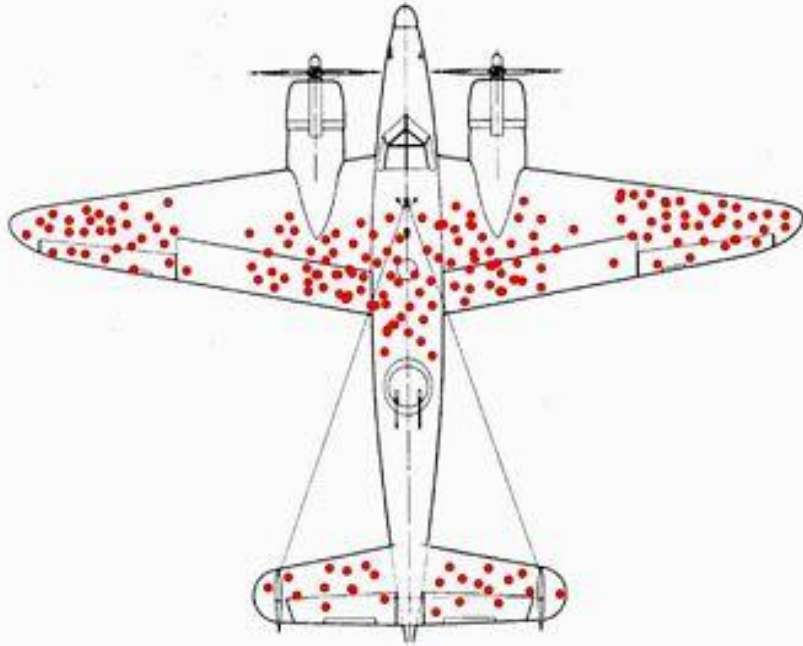


“Exponential Technical Debt”

- **1990s**
 - Automated Operations
- **2000s**
 - Automated Infrastructure
- **2010s**
 - Composable Infrastructure
- **2020s**
 - Generative AI



Looking for Where the Bullet Holes Are



“What Abraham Wald found was a logical error known as Survival Bias”

Dear Auditor,



a love letter to auditors from devops,
where we promise to make life better

[View My GitHub Profile](#)

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ZIP File

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TAR Ball

View On
GitHub

Hosted on GitHub Pages — Theme by [orderedlist](#)

Dear Auditor,

We realize that we have been changing things in a rapid fashion from Agile and DevOps to Cloud and Containers. Yes, we have been busy, and are having great success delivering faster than ever, with better quality and supporting the business response to competitive pressures. This isn't just icing on the cake, the only sustainable advantage in our industries is the ability to meet customer demands faster, more reliably than our competitors.

With all this growth, we made a mistake, we forgot to bring you along for the ride. That is totally our bad, but we want to make it right. We want to make some new commitments.

- We will bring you along
- We will be fully transparent about our development process
- We do realize that we own the risks
- We will maintain an open channel of discussion to demonstrate to you how we manage risks with our modern development practices

For example, you have told us that you are concerned about “Separation of Duties” in agile and DevOps practices, and we heard you! We think we have a better way to manage this and risks now. Having everything in version control, enforcing peer review for every change, releasing via a secure pipeline, restricting production access, and monitoring unauthorized changes in production systems should address your concern.

We Need To Think Differently

Industry Working Groups



AN UNLIKELY UNION: DEVOPS AND AUDIT

October 4, 2017



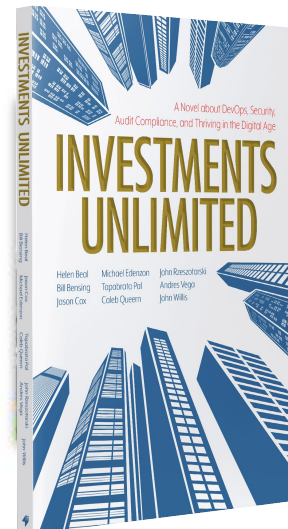
DEVOPS AUTOMATED GOVERNANCE REFERENCE ARCHITECTURE

September 10, 2019



DEAR AUDITOR

August 01, 2018



Shadow AI is inevitable

The Rise of Shadow AI

Published On: July 5, 2023 By John Willis



"Before AI is smarter than us, I think the people developing it should be encouraged to put a lot of work into understanding how it might try and take control away," – Geoffrey Hinton, Godfather of A.I.

- The rise of Generative AI and ChatGPT is leading to a new version of Shadow IT with increased security and compliance risks.
- Generative AI large language models are complex and opaque, with few experts understanding how they work, leading to increased security risks.
- Organizations must learn to manage the increased risks of using open source software in generative AI solutions.
- IT leaders need to implement standardized internal training programs, standardize the use of LLMs and develop secure LLM delivery workflows.
- New policies and observability patterns must be developed to consider vulnerability scans of LLM code and dependencies and detect anomalies, breaches and data leaks.
- Recent innovations like retrieval augmentation tools have reduced LLM hallucinations and allowed for safer

It's going to happen

- People avoiding bottlenecks
- Enthusiasm + Need

How do we *safely* harness it?

**Insanity is doing
the same thing over
and over again and
expecting different
results**

Rita Mae Brown



Dear CIO,

We are writing to inform you about a significant emerging risk that every organization will soon face: Shadow AI. Just as Shadow IT posed challenges in the past, the uncontrolled proliferation of AI technologies across organizations could lead to even greater risks if not addressed proactively. As AI becomes more accessible, business units deploy their AI solutions without proper governance, security, or oversight. While this can drive innovation, it also introduces critical risks:

- 1. Security vulnerabilities and data exposure**
- 2. Inconsistent governance and compliance**
- 3. Technical debt and interoperability issues**
- 4. Cost inefficiencies and wasted resources**
- 5. Repetitional damage from AI missteps**

Who is Deming



Deming Quote Wall

“Quality is everyone’s responsibility.”

“Learning is not compulsory... neither is survival.”

“In God we trust; all others bring data.”

“A bad system will beat a good person every time.”

“Every system is perfectly designed to get the result that it does.”



Why Deming Matters



In 2011 Marc Andreessen said

**"Software Is Eating
the World"**

In 2015 Josh Corman said

**"Software Is Infecting
the World"**

In 2022 John Willis and Josh Corman said

**"Software is Screwing
with Maslow"**

(Health, Food, Water, Shelter)

In 2023 John Willis says

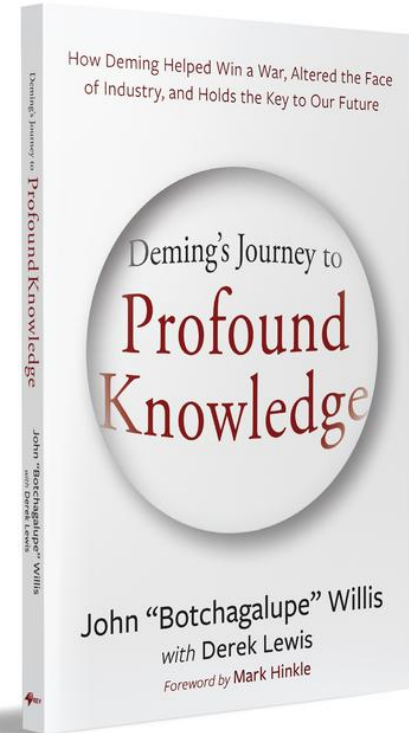
**"Generative AI
is ????????**
Everything

**Deming and
the “Cyber”
Future World**



Deming Management Philosophy

- Operational Definitions
- System of Profound Knowledge
- Fourteen Points for Management



Operational Definition



An operational definition is a procedure agreed upon for translation of a concept into measurement of some kind



New Economics



**There is no true value of any
characteristic, state, or condition
that is defined in terms of
measurement or observation**



The New Economics
Deming, W. Edwards

Count the Number of Horses, Elephants, and Pigs?



Count the Number People in a Restaurant?



Operationalism



To find the length of an object, we have to perform certain physical operations. The concept of length is therefore fixed when the operations by which length is measured are fixed. The concept of length involves as much as and nothing more than the set of operations by which length is determined



Percy Williams Bridgman

System of Profound Knowledge (SoPK)

Decision

Whether the object or material met the criteria. Test results are used to determine if a characteristic meets a criterion. Clarify a documented statement.

In line with PDSA's "Study" portion. Analytical statistics via SPC emphasizes that Special Cause variation is the starting point of a required "Act."



Criteria

Standards against which to evaluate test results. Provide judgment criteria. The "Plan" portion of PDSA aligns with this. A Theory of Knowledge (how do we know what we think we know) is also an element of profound knowledge.

Test

Measuring a characteristic. How is clarity determined, and who performs the test?

This aligns with the "Do" portion of PDSA. Furthermore, it aligns with the Theory of Variation element of profound knowledge.

“Words Matter” In DevOps

- **Time**
- **Root**
- **Zero**
- **Hermetic**
- **Service**

“Words Matter” AI

- **Foundational**
- **Frontier**
- **Large**
- **Small**
- **Open**
- **Hallucinations**

Operational Definitions of DORA Metrics

Lead Time

Time to Restore

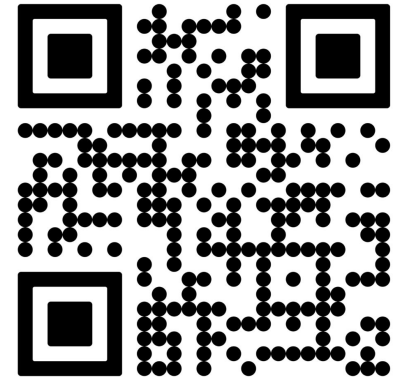
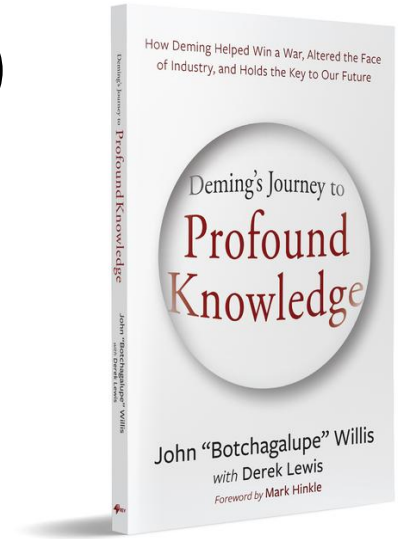
What is Lead Time, or Time to Restore?

- **Criteria:** Why are we asking? What is good or bad, or standard? Are all the teams in an org aligned on the same operational definition? Industry?
- **Test:** What's the measurement? Delivery or deployment, branch, master branch or pull request? Dark launch, feature flag. Mean, mode, standard deviation?
- **Decision:** Did it meet the criteria? What action do we take? Are we continuously improving?

System of Profound Knowledge (SoPK)

Four Elements as a Lens for understanding Complexity

- **Theory of Knowledge**
 - How do we know?
- **Theory of Variation**
 - How do we understand?
- **Theory of Psychology**
 - How are people motivated?
- **Appreciation for a Systems**
 - Systems thinking integrates all four.



14 Points in a Cyber World

1. Create constancy of purpose
2. Adopt the new philosophy
3. Cease dependence on inspection to achieve quality
4. End the practice of awarding business on the basis of price tag
5. Improve constantly and forever the system of production and service
6. Institute training on the job
7. Institute leadership
8. Drive out fear, so that everyone may work effectively for the company
9. Break down barriers between departments
10. Eliminate slogans, exhortations, and targets
11. Eliminate work standards (quotas) and management by objective
12. Remove barriers that rob the hourly worker of his right to pride of workmanship
13. Institute a vigorous program of education and self-improvement.
14. Put everybody in the company to work to accomplish the transformation

CIO's Need To Get A Handle On GAI.

#1 Create Purpose

#2 Adopt a New Philosophy

#5 Never Stop Improving

This is a never ending cat-and-mouse game. The bad guys do bad things while the good guys try to keep up as well as they can. Shannon Lietz takes what I call a uniformity approach. Instead of a purely defensive stance, she takes a proactive stance which she calls adversary management. She monitors the behavior and motivations of attackers. How often do adversaries attack? What's their purpose or aim? What do they target and what do they ignore? How long will they mount an attack before giving up? In effect,

#7 Don't Manage—Lead

Most organizations have enough resources to resolve only P1 incidents. Managers responsible for cybersecurity are primarily motivated by reducing P1 incidents. The more they see that number shrink, the better they believe things are. Incidents are software quality issues that need to be dealt with quickly and quietly. This is not leadership. This is firefighting. Ed, in contrast, would get rid of the arbitrary four categories and instead use SPC to create a mathematical approach to managing security incidents.

Profound - John Willis

#5 Never Stop Improving



#9 No Silos

The US House of Representatives committee investigating the breach found some institutional organizational barriers between the chief information security officer (CISO) and the chief information officer (CIO): the CISO didn't report to the CIO, as you might assume with both of them being responsible for the company's tech. She didn't even report to the CEO. The person responsible for cybersecurity of multibillion-dollar corporation reported to...a lawyer.

Profound - John Willis

#11 Quality, Not Quotas

A goal that lies beyond the means of its accomplishment will lead to discouragement, frustration, demoralization. In other words, there must be a method to achieve an aim. By what method?

Deming, W. Edwards. The New Economics for Industry, Government, Education, third edition .

MBO's, MBR's, OKR's

The required transformation of Western style of management requires that managers be leaders. Focus on outcome (management by numbers, MBO, work standards, meet specifications, zero defects, appraisal of performance) must be abolished, leadership put in place.

Deming, W. Edwards. Out of the Crisis (MIT Press) (p. 54).

Management by Means

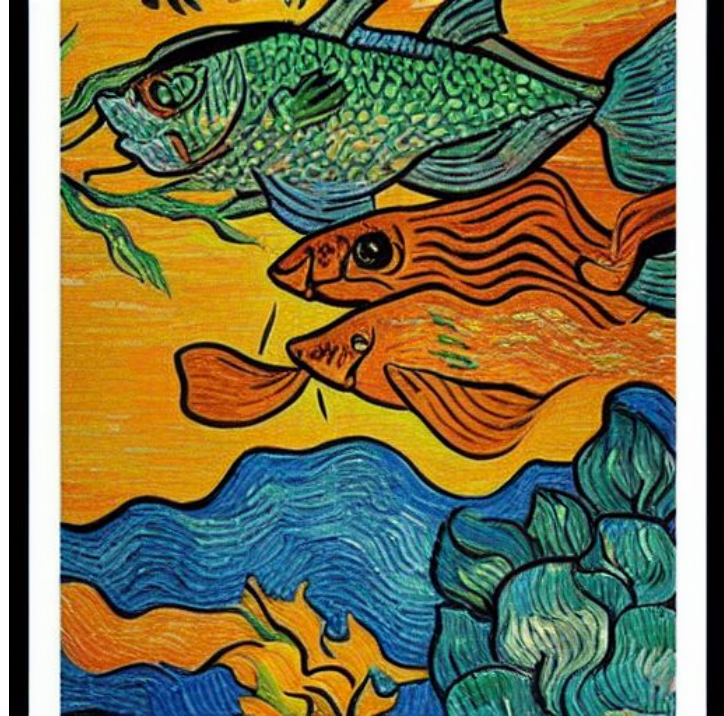
If you can accomplish a goal without a method, then why were you not doing it last year?

Deming, W. Edwards. The New Economics for Industry, Government, Education, third edition - (Lloyd Nelson)

In Other Words

**Give a “person” a Fish, and
You Feed “them” for a Day.
Teach a “person” To Fish,
and You Feed “Them” for a
Lifetime**

Chinese Proverb?

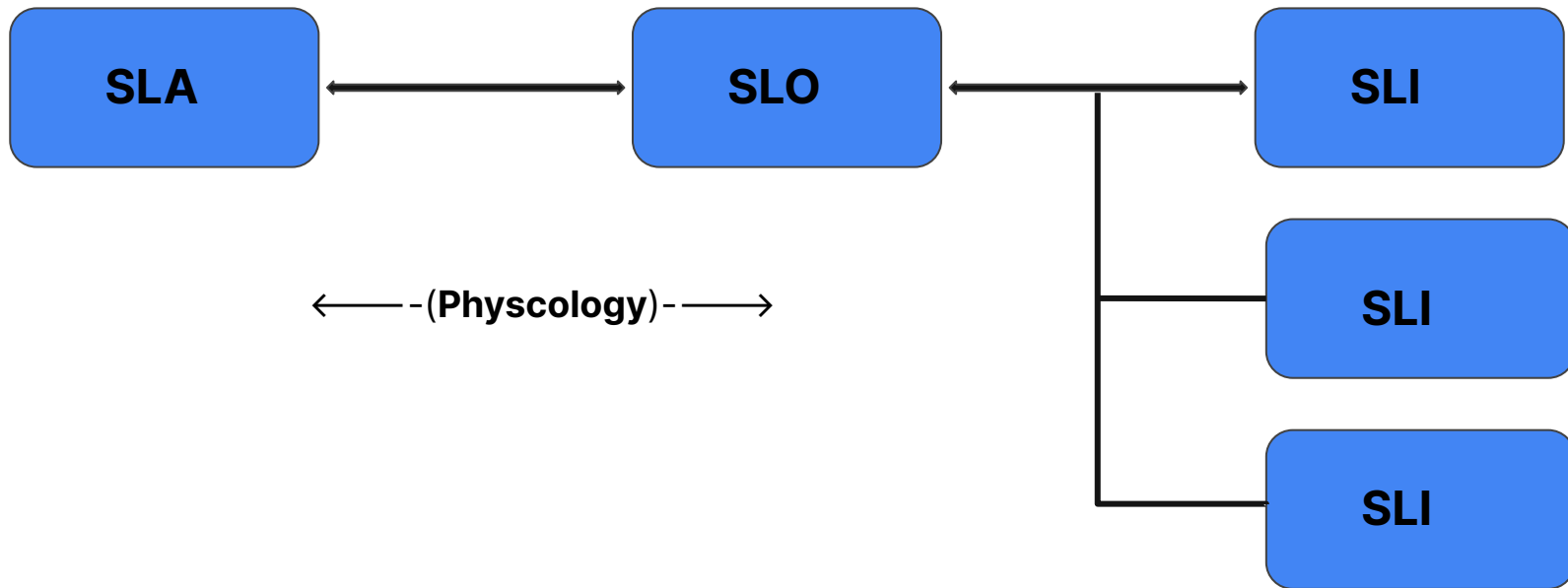


SRE is Profound Knowledge

Service Level Agreement
Between provider and user
(**Systems**)

Service Level Objective
A services objective
(**Variation**)

Service Level Indicator
Performance indicators
(**Knowledge**)



#12 No Barriers to Pride

I've seen this over and over again in the tech industry. Take one metric for example: KLOC, or “defect per thousand lines of code.” One KLOC is considered good: Out of a thousand lines of code, there was only one bug or error. From a deterministic perspective, it's logical to want that number to be as close to zero as possible—100% clean code. Therefore, you would always want that number to be going down.

Profound - John Willis

Knight Capital (API Anomaly Detection)

In 2012, Knight was the largest trader of stocks on the NYSE and Nasdaq. Knight's Electronic Trading Group managed an average of 3.3 billion trades each day, worth over \$21 billion.

440 Million in 45 minutes.

Profound Knowledge - SEC Cease and Desist

- **Knowledge**

- Second technician to review the code installation.

- **Variation**

- Insufficient to ensure an orderly installation of new code nor to prevent the activation of code no longer intended for use.

- **Physcology**

- Reasonably designed to guide employees' responses to significant technological and compliance incidents.

- **Systems**

- Adequate written description of its risk management controls.

Why Now



In 2023 Botchagalupe says

**"Generative AI
is ????????
Everything**

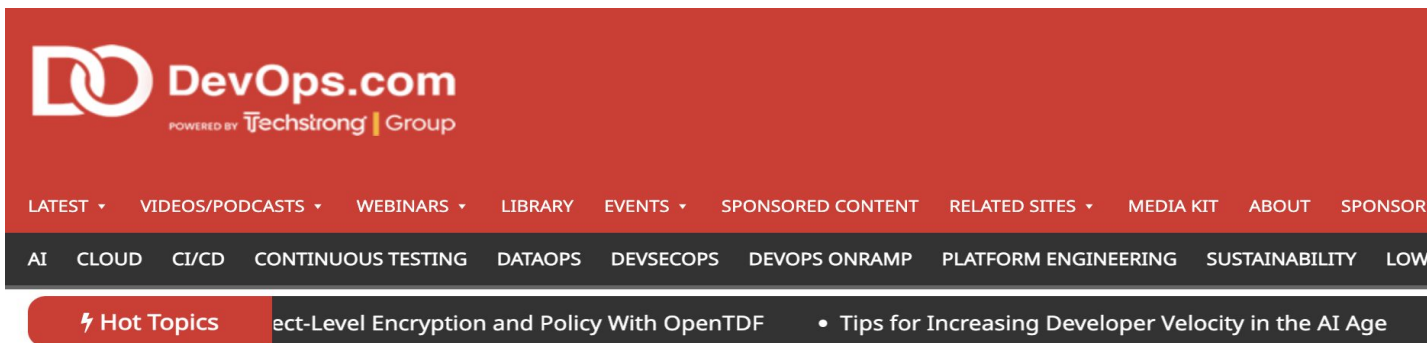
Technical Debt



The Air Canada Chatbot Incident: A Cautionary Tale



I Predicted Both “Dark GAI” and “GAI Squatting”



[Blogs](#) [What DevOps Needs to Know About ChatGPT](#)

What DevOps Needs to Know About ChatGPT



BY: [JOHN WILLIS](#) ON JANUARY 24, 2023 — [2 COMMENTS](#)

OpenAI's [generative pre-trained transformers \(GPTs\)](#) bring powerful AI language models to every IT community. Within the past few months, [ChatGPT](#) has become an incredibly popular derivative of GPT. GPT also has the potential to shake up the DevOps and DevSecOps communities. This article will provide an overview of GPT, its history, current status and future effects on IT, especially DevOps.

GenAI Basics - LORMA Stack

- **L** - Language Model Orchestration (LLM/SLM)
- **O** - Observability
- **R** - Retrieval Augmentation Generation
- **M** - Model Providers
- **A** - Agentic Agent Management

OWASP



LLM AI Cybersecurity & Governance Checklist

From the OWASP Top 10
for LLM Applications Team

Version: 1.1

Published: April 11, 2024

AI at Work Is Here. Now Comes the Hard Part



Finding 1

Employees want AI at work—and won't wait for companies to catch up

They're bringing their own tools even as leaders face AI inertia.

-
- **75%** of knowledge workers around the world use generative AI at work.
 - **78%** of AI users are bringing their own AI to work (BYOAI).
 - While **79%** of leaders believe their company needs to adopt AI to stay competitive, **60%** of leaders worry their organization's leadership lacks a plan and vision to implement it.



Finding 2

For employees, AI raises the bar and breaks the career ceiling

Some are itching for a career change, and there is a massive opportunity for those willing to skill up on AI.

-
- **66%** of leaders say they would not hire someone without AI skills.
 - **71%** say they'd rather hire a less experienced candidate with AI skills than a more experienced candidate without.
 - There was a **142x** increase in skills like Copilot and ChatGPT added to LinkedIn profiles last year.



Finding 3

The rise of the AI power user—and what they reveal about the future

Power users use AI at least several times per week. They say it saves them more than 30 minutes per day.

-
- Frequently experimenting with AI is the **#1** predictor of an AI power user.
 - Power users say AI boosts their creativity (**92%**) and helps them focus on the most important work (**93%**).
 - AI also helps them feel more motivated (**91%**) and enjoy work more (**91%**).

Wiz Research finds architecture risks that may compromise AI-as-a-Service providers and consequently risk customer data; works with Hugging Face on mitigations

Wiz researchers discovered architecture risks that may compromise AI-as-a-Service providers and put customer data at risk. Wiz and Hugging Face worked together to mitigate the issue.



Shir Tamari, Sagi Tzadik
April 4, 2024

12 minutes read



Agentic DevOps



[Home](#) » [Videos](#) » Copilot to No Pilot: The Rise of Autonomous DevTools with Ray Myers at AIE 2024

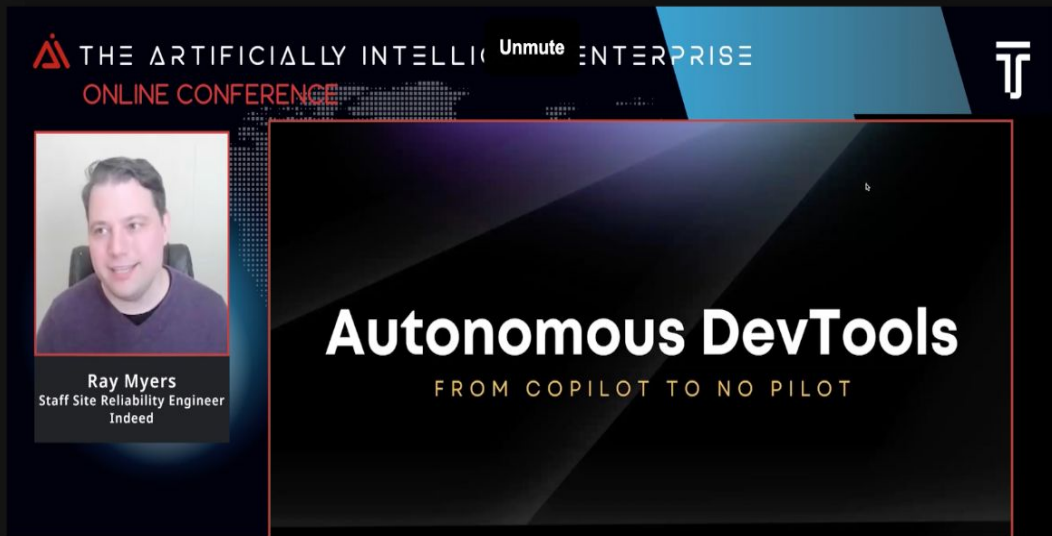
Copilot to No Pilot: The Rise of Autonomous DevTools with Ray Myers at AIE 2024

July 3, 2024

Earlier this year, Cognition released the viral demo Devin, marking a new generation of **AI** coding tools with unprecedented autonomy. Open Source competitors quickly followed. While previous “assistants” operated simply as an auto-complete in an editor, these new tools promise to carry entire tasks through to completion! Will this save us from an avalanche of tech debt, or just create a new one?

Share some ❤

Host: [Ray Myers](#)  Categories: [AIE 2024](#)



SWE-bench



Can Language Models Resolve Real-World GitHub Issues?

ICLR 2024

Carlos E. Jimenez*, John Yang*,
Alexander Wettig, Shunyu Yao, Kexin Pei,
Ofir Press, Karthik Narasimhan

 Paper

 Code

 Submit

 Analysis

🔥 Evaluating on SWE-bench just became a lot more reliable! SWE-bench evaluation now uses **Docker** for easier, containerized, reproducible evaluation. [\[Report\]](#)

Leaderboard

Model	% Resolved	Date	Logs	Trajs	Site	Verified?	Open?
🏆 Factory Code Droid	19.27	2024-06-17		-		✗	✗
🥈 AutoCodeRover (v20240620) + GPT 4o (2024-05-13)	18.83	2024-06-28		-		✗	✗
🥉 AppMap Navie + GPT 4o (2024-05-13)	14.60	2024-06-15		-		✓	✓
Amazon Q Developer Agent (v20240430-dev)	13.82	2024-05-09		-		✗	✗
SWE-agent + GPT 4 (1106)	12.47	2024-04-02				✓	✓
SWE-agent + Claude 3 Opus	10.51	2024-04-02			-	✓	✓

gptscript-ai / gptscript main

Model config #552

Need a model config to specify max tool count, context length, and other model specific constraints.

related issue: <https://github.com/gptscript-ai/gptscript/issues/547>



Specification



Plan



Implementation ⓘ



5 files changed



Open file explorer



Open terminal



Open codespace

[Create pull request](#)

Create pull request



Branch name *

model-config

Title *

Model config

Description

Related to #552

Add model-specific constraints configuration.

***Documentation**:

 Add a new section in `docs/docs/03-tools/05-context.md` to explain model-specific constraints, including examples for max tool count and context length.

***Builtin Tools**:

 Update `pkg/builtin/builtin.go` to include model-specific constraints[Create pull request](#)

488

489 func (c *Client) call(ctx context.Context, request o

492

493 func (c *Client) call(ctx context.Context, request o

amResponse := os.Getenv("GPTSCRIPT_INTERNAL_

al <- types.CompletionStatus{

CompletionID: transactionID,

PartialResponse: &types.CompletionMessage{

Role: types.CompletionMessageRoleType

Content: types.Text("Waiting for model r

Debug("calling openai", "message", request.

streamResponse {

request.StreamOptions = nil

resp, err := c.c.CreateChatCompletion(ctx, r

if err != nil {

return nil, err

return []openai.ChatCompletionStreamResponse

{

ID: resp.ID,

Object: resp.Object,

Created: resp.Created,

Model: resp.Model,

Usage: resp.Usage,

Choices: []openai.ChatCompletionStre

{

Index: resp.Choices[0].Index

514

Usage: resp.Usage,

516

Choices: []openai.ChatCompletionStre

518

{

520

Index: resp.Choices[0].Index

521

GenAI Threats



GenAI Threats (Not using GenAI)

- Competitive Disadvantage
- Market Perception
- Innovation Stagnation
- Operational Inefficiencies
- Inefficient Allocation of Human Resources

GenAI Threats (Using GenAI)

- Immaturity of OVAL, STIX, CVE, and CWE
- New Malware Opportunities
- Malicious Pickle-Serialized Model
- Zero-Day Embeddings Vulnerabilities
- Phishing Schemes
- Reverse Engineering
- Innovating Hacking Techniques
- Deep Fakes

OWASP Top 10 for LLM Applications

LLM01

Prompt Injection

This manipulates a large language model (LLM) through crafty inputs, causing unintended actions by the LLM. Direct injections overwrite system prompts, while indirect ones manipulate inputs from external sources.

LLM02

Insecure Output Handling

This vulnerability occurs when an LLM output is accepted without scrutiny, exposing backend systems. Misuse may lead to severe consequences like XSS, CSRF, SSRF, privilege escalation, or remote code execution.

LLM03

Training Data Poisoning

This occurs when LLM training data is tampered, introducing vulnerabilities or biases that compromise security, effectiveness, or ethical behavior. Sources include Common Crawl, WebText, OpenWebText, & books.

LLM04

Model Denial of Service

Attackers cause resource-heavy operations on LLMs, leading to service degradation or high costs. The vulnerability is magnified due to the resource-intensive nature of LLMs and unpredictability of user inputs.

LLM05

Supply Chain Vulnerabilities

LLM application lifecycle can be compromised by vulnerable components or services, leading to security attacks. Using third-party datasets, pre-trained models, and plugins can add vulnerabilities.

LLM06

Sensitive Information Disclosure

LLMs may inadvertently reveal confidential data in its responses, leading to unauthorized data access, privacy violations, and security breaches. It's crucial to implement data sanitization and strict user policies to mitigate this.

LLM07

Insecure Plugin Design

LLM plugins can have insecure inputs and insufficient access control. This lack of application control makes them easier to exploit and can result in consequences like remote code execution.

LLM08

Excessive Agency

LLM-based systems may undertake actions leading to unintended consequences. The issue arises from excessive functionality, permissions, or autonomy granted to the LLM-based systems.

LLM09

Overreliance

Systems or people overly depending on LLMs without oversight may face misinformation, miscommunication, legal issues, and security vulnerabilities due to incorrect or inappropriate content generated by LLMs.

LLM10

Model Theft

This involves unauthorized access, copying, or exfiltration of proprietary LLM models. The impact includes economic losses, compromised competitive advantage, and potential access to sensitive information.

How?



Think Differently

- *Erotetics* is crucial for framing AI inquiries and outputs.
- Well-defined language are essential for accurate AI responses.
- Operational definitions ensure shared understanding and reliable AI outputs.
- Precision in prompts and definitions affects AI quality and guards against errors.
- As AI use grows, systematic question-framing and definitions are paramount.
- Applying rigor in erotetics and definitions optimizes AI effectiveness.

Constraints Enable Freedom

By curtailing the potential variation in component behavior... context-dependent constraints paradoxically also create new freedoms for the overall system



Alicia Juarrero

Ashby's Law

If a system is to be stable the number of states of its control mechanism must be greater than or equal to the number of states in the system being controlled



Ross Ashby

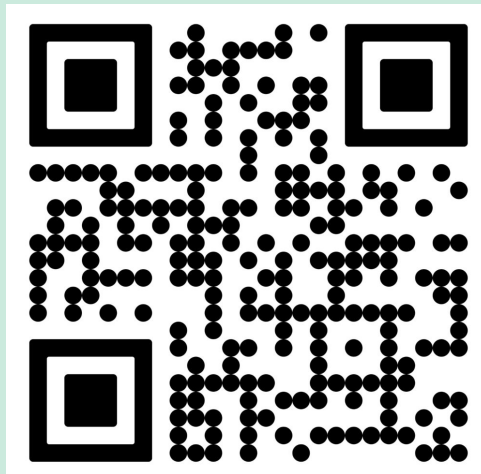
Suggestions for Generative AI

- Legacy Migration Assessment
- Security and Privacy Training for GenAI
- Platform Engineering for GenAI
- Service Reliability Engineering for GenAI
- Secure Supply Chain for GenAI
- Automated Governance for GenAI

Thank you

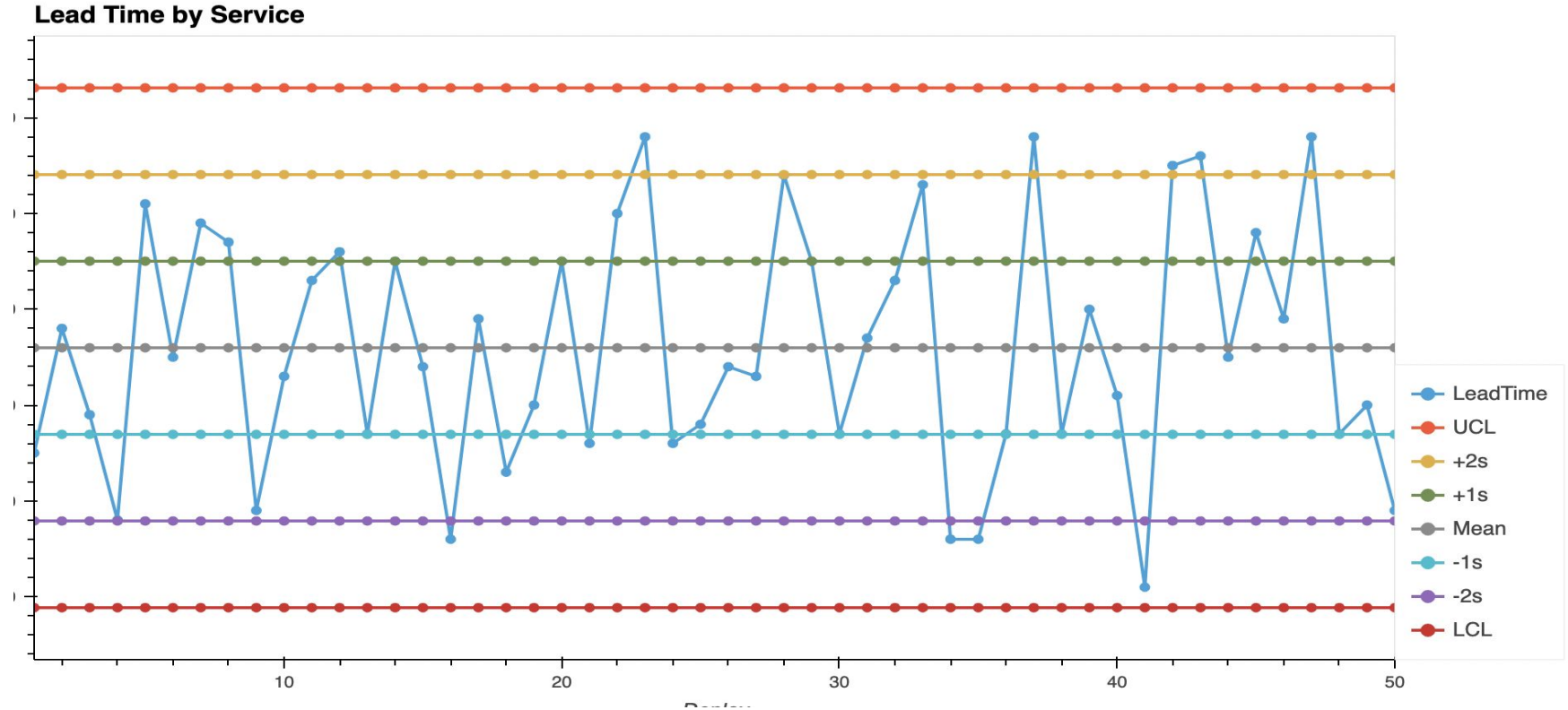
Find out more at...

- [Profound-deming.com](https://profound-deming.com)
- linkedin.com/in/johnwillisatlanta/
- [@botchagalupe](https://twitter.com/botchagalupe)
- botchagalupe@gmail.com



Bonus Material

Analytical Statistics



SoPK for IT Risk

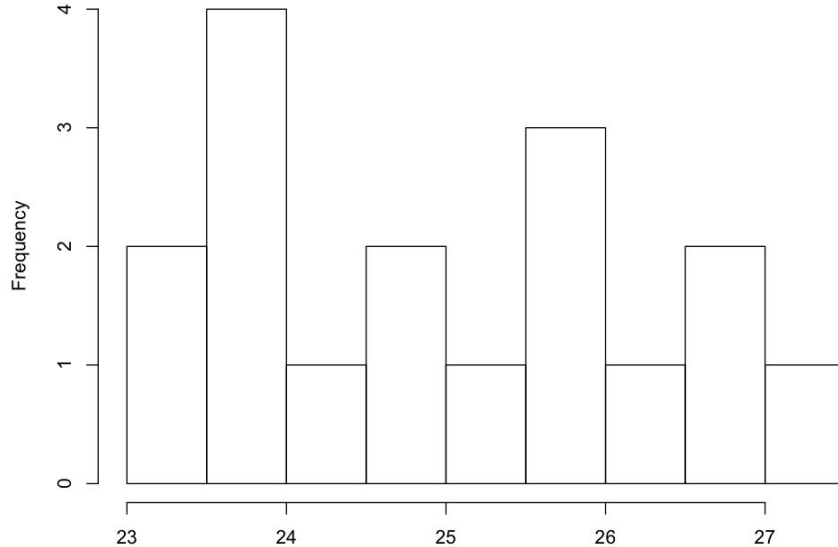
- **Risk Control: Container Security Scanning**
 - Attestation: “All dependencies in this build satisfy any known vulnerables.”
 - Pass/Fail
- **Risk Control: Unit tests**
 - Attestation: “All tests executed and passed.”
 - Percentage
- **Risk Control: Information Leakage**
 - Attestation: “Secret information observable information defects.”
 - Pass/Fail

SoPK for IT Risk

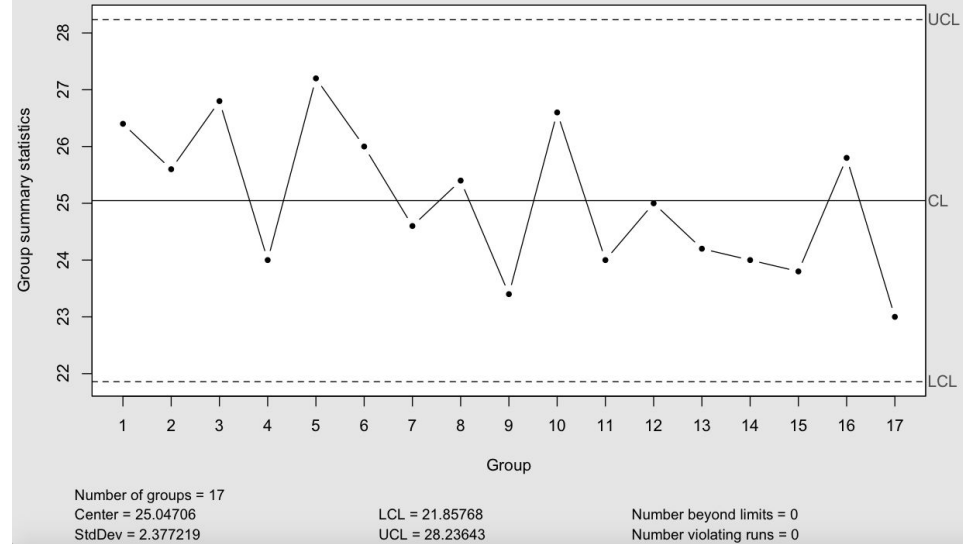
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Enumerated Versus Analytical

Histogram of rowMeans(SCA[1:17,])

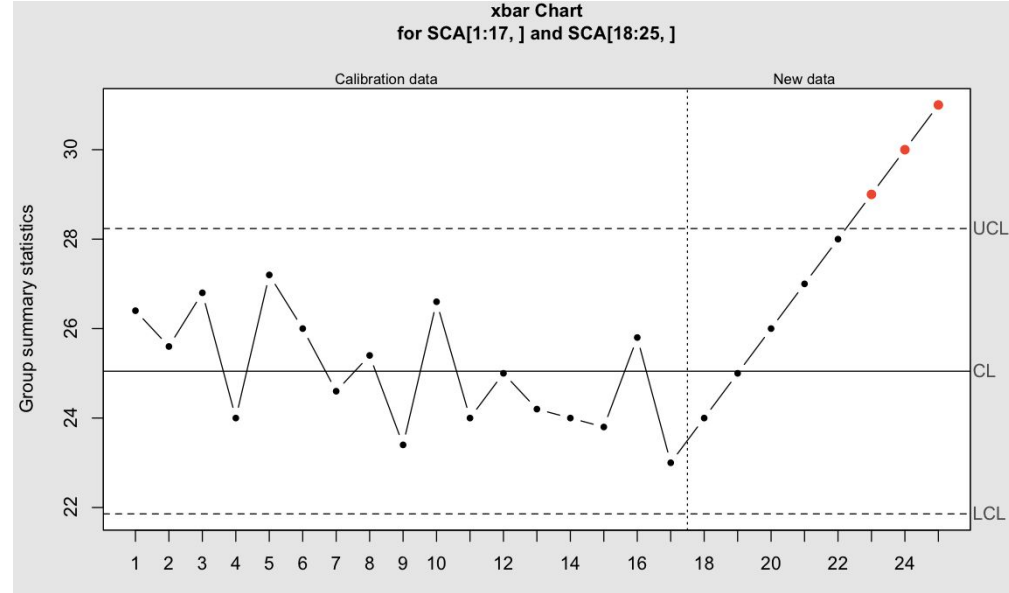
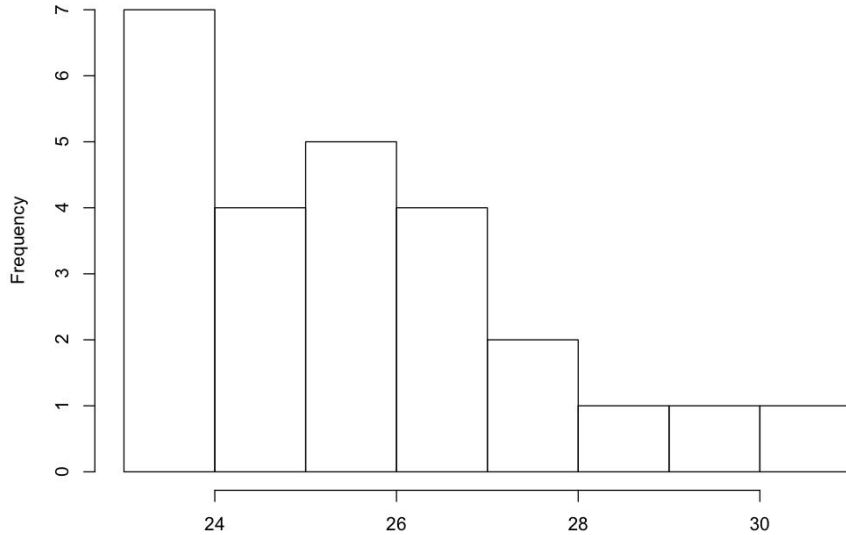


xbar Chart
for SCA[1:17,]



Enumerated Versus Analytical

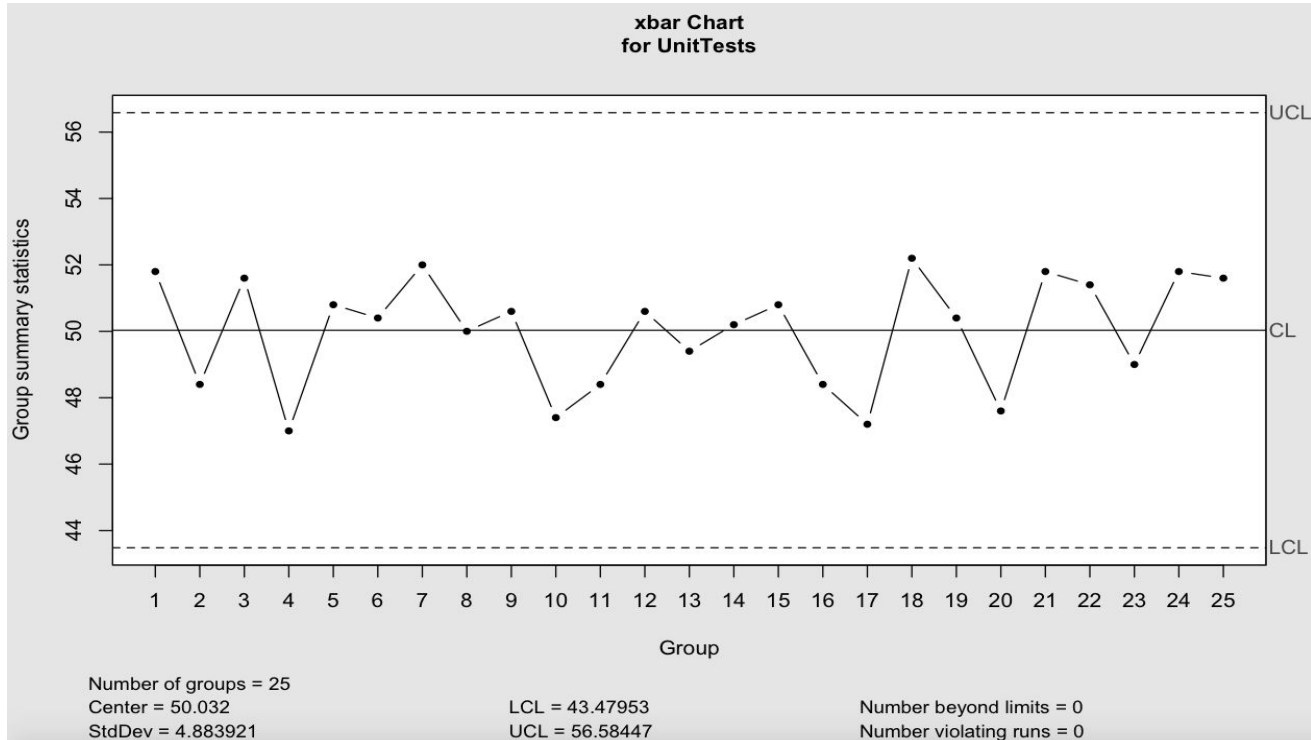
Histogram of rowMeans(SCA)



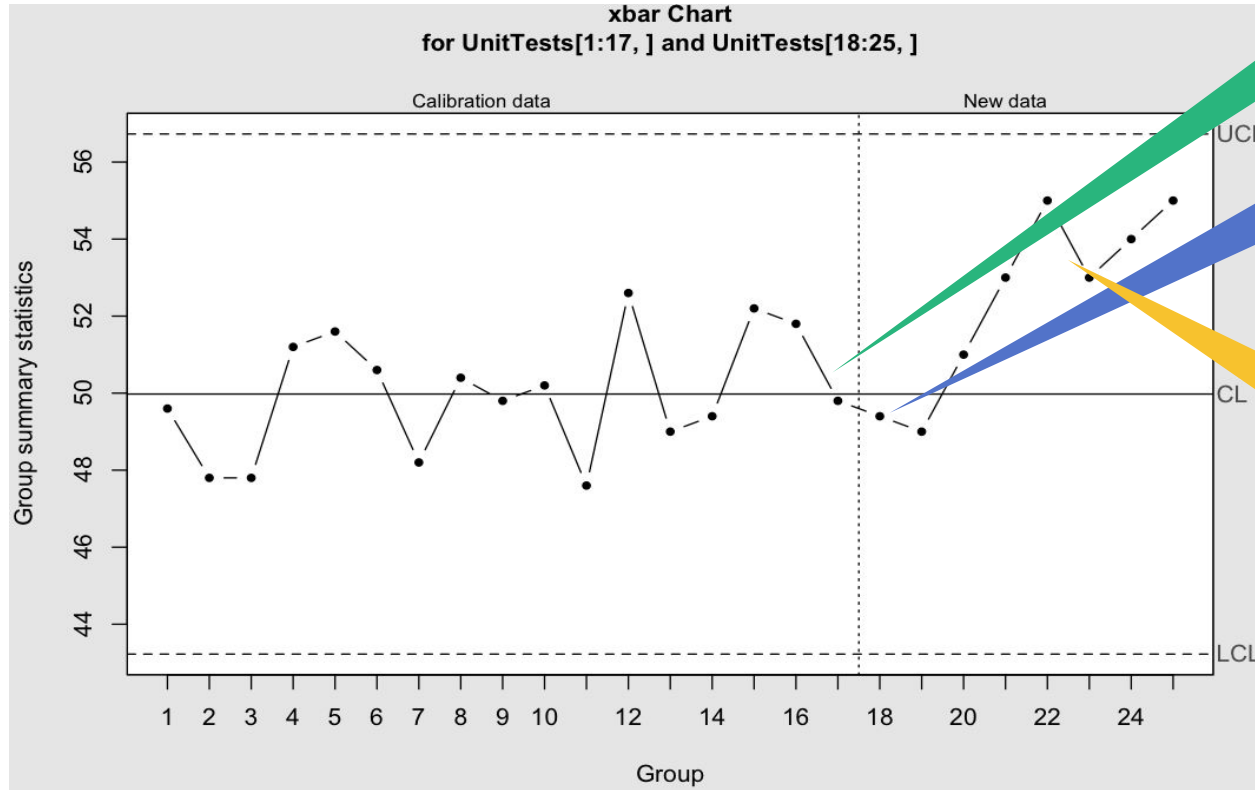
SoPK for IT Risk

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SoPK for IT Risk



SoPK for IT Risk



PLAN

DO

STUDY

The new normal

ACT

